

The local scalar-loop counterterm in de Sitter space

Revised 10 September 2026

Abstract

The proper-time calculation proposed in the project notes determines the local ultraviolet part of the scalar vacuum polarization. For one charged complex scalar, the coefficient is $\partial\delta_3/\partial\ln\Lambda^2 = -e^2/(48\pi^2)$. We derive it with a proper-time cutoff and with dimensional regularization, keeping the charge and effective-action conventions explicit. The exponential factor in the original calculation is an invariant four-volume and cancels in the matching. The finite response at the curvature scale requires the nonlocal kernel; its separation from a local counterterm depends on the chosen renormalization condition.

1 Operator and effective-action convention

We consider the same charged complex scalar as in Step 1, with

$$M_\phi^2 = m_\phi^2 + \xi R = m_\phi^2 + 12\xi H^2, \quad D_\mu = \nabla_\mu - ieA_\mu. \quad (1)$$

The Lorentzian metric has signature $(-+++)$. The photon kinetic term and its counterterm are

$$S_{F,L} = -\frac{1}{4} \int d^4x \sqrt{-g} (1 + \delta_3) F_{\mu\nu} F^{\mu\nu}. \quad (2)$$

For the heat kernel we continue to Euclidean signature and use the conventional effective action $\Gamma_E = -\ln Z_E$. Gaussian integration gives

$$P = -D_E^2 + m_\phi^2 + \xi R, \quad Z_\phi = (\det P)^{-1}, \quad \Gamma_{E,\phi}^{(1)} = \text{Tr} \ln P. \quad (3)$$

There is no factor of $1/2$, since the scalar is complex. The Euclidean tree action is $+\frac{1}{4} \int \sqrt{g} F_E^2$. These signs are convenient because both the determinant and the counterterm can be matched directly in Euclidean signature. References which define $\Gamma = \ln Z$, including the proper-time convention used in the project outline, reverse the action signs together and give the same δ_3 .

On de Sitter, R is constant. Factoring it into the mass is therefore an exact rearrangement of the operator:

$$\Gamma_{E,\phi}^{(1)} = - \int_0^\infty \frac{dT}{T} e^{-M_\phi^2 T} \text{Tr} e^{TD_E^2}, \quad (4)$$

up to an additive field-independent normalization. The local expansion below concerns the bulk coefficient. A finite time interval will only be used to evaluate its integral; it is not a replacement of the determinant by a boundary-value problem on a manifold with physical boundaries.

2 The electromagnetic heat-kernel coefficient

In d Euclidean dimensions,

$$\mathrm{Tr} e^{TD_E^2} \sim \frac{1}{(4\pi T)^{d/2}} \int d^d x \sqrt{g} \sum_{n=0}^{\infty} b_n(x) T^n, \quad T \rightarrow 0^+. \quad (5)$$

The connection curvature is

$$\Omega_{\mu\nu} = [D_\mu, D_\nu] = -ieF_{\mu\nu}, \quad b_2|_{F^2} = \frac{1}{12} \Omega_{\mu\nu} \Omega^{\mu\nu} = -\frac{e^2}{12} F_{E,\mu\nu} F_E^{\mu\nu}. \quad (6)$$

This is the coefficient quoted by Bastianelli, Dávila and Schubert [3], with the electric charge restored. In the mass-dimension notation of Vassilevich [4], the same term belongs to a_4 . Here the index labels the power of T .

Neither $b_0 = 1$ nor $b_1 = R/6$ contains F^2 . Pure gravitational terms in b_2 renormalize gravitational couplings and can be kept separate. If instead one leaves $e^{-m_\phi^2 T}$ outside the heat kernel, its first coefficient is $a_1 = (1/6 - \xi)R$. The two organizations obey

$$\sum_{n=0}^{\infty} a_n T^n = e^{-\xi R T} \sum_{n=0}^{\infty} b_n T^n. \quad (7)$$

The curvature coupling is counted once in either organization.

The b_2 contribution to (4) is

$$\Gamma_{E,F^2}^{(1)} = \frac{e^2}{192\pi^2} I(M_\phi^2) \int d^4 x \sqrt{g} F_E^2, \quad I(M_\phi^2) = \int_0^\infty \frac{dT}{T} e^{-M_\phi^2 T}. \quad (8)$$

The integral is ultraviolet divergent. Extending a single small- T coefficient to all proper times, as in (8), also assigns it a finite mass-dependent logarithm. That finite assignment is useful for matching schemes, but the exact finite determinant is not obtained by this truncation.

3 Two regulators

3.1 Proper-time cutoff

For $\mathrm{Re} M_\phi^2 > 0$, impose $T \geq \Lambda^{-2}$. With $u = M_\phi^2 T$,

$$I_\Lambda(M_\phi^2) = \int_{\Lambda^{-2}}^\infty \frac{dT}{T} e^{-M_\phi^2 T} = \Gamma\left(0, \frac{M_\phi^2}{\Lambda^2}\right), \quad (9)$$

$$= \ln \frac{\Lambda^2}{M_\phi^2} - \gamma_E + O\left(\frac{M_\phi^2}{\Lambda^2}\right). \quad (10)$$

Thus the unintegrated expression $-\gamma_E - \ln(m_\phi^2 T)$ in the project outline is to be understood as a lower-endpoint expansion of a regulated integral. No proper-time variable remains in the effective action.

The restriction on M_ϕ^2 concerns the large- T end: zero mass gives an infrared divergence and a negative real mass squared gives an unstable Euclidean scalar operator. These issues do not change the coefficient of the small- T ultraviolet singularity.

3.2 Dimensional regularization

Let $d = 4 - 2\epsilon$ and introduce μ_{DR} . The required factor is

$$\frac{\mu_{\text{DR}}^{2\epsilon}}{(4\pi)^{2-\epsilon}} \int_0^\infty dT T^{\epsilon-1} e^{-M_\phi^2 T} = \frac{1}{(4\pi)^2} \left[\frac{1}{\epsilon} + \ln \frac{\mu_{\text{DR}}^2}{M_\phi^2} + O(\epsilon) \right], \quad (11)$$

$$\frac{1}{\epsilon} = \frac{1}{\epsilon} - \gamma_E + \ln 4\pi. \quad (12)$$

Consequently,

$$\Gamma_{E,F^2}^{(1)} = \frac{e^2}{192\pi^2} \left[\frac{1}{\epsilon} + \ln \frac{\mu_{\text{DR}}^2}{M_\phi^2} \right] \int d^4x \sqrt{g} F_E^2 + O(\epsilon). \quad (13)$$

Here M_ϕ^2 denotes its four-dimensional value. Continuing it as $m_\phi^2 + \xi d(d-1)H^2 = M_\phi^2 + O(\epsilon)$ changes $(M_\phi^2)^{-\epsilon}$ only at $O(\epsilon^2)$. After multiplication by $\Gamma(\epsilon)$, that change is $O(\epsilon)$; it affects neither the pole nor the finite term displayed in (13).

4 Matching the field-strength counterterm

The Euclidean counterterm is $\Gamma_{E,\text{ct}} = \delta_3 \int \sqrt{g} F_E^2/4$. Cancelling the logarithmic part of (8) therefore gives

$$\frac{\delta_{3,\log}}{4} + \frac{e^2}{192\pi^2} \ln \frac{\Lambda^2}{M_\phi^2} = 0, \quad \delta_{3,\log} = -\frac{e^2}{48\pi^2} \ln \frac{\Lambda^2}{M_\phi^2}. \quad (14)$$

The regulator-independent statement is

$$\boxed{\frac{\partial \delta_3}{\partial \ln \Lambda^2} = -\frac{e^2}{48\pi^2}}. \quad (15)$$

In $\overline{\text{MS}}$, the corresponding equations are

$$\delta_{3,\overline{\text{MS}}} = -\frac{e^2}{48\pi^2\epsilon}, \quad \Gamma_{E,F^2,\overline{\text{MS}}}^{(1)} = \frac{e^2}{192\pi^2} \ln \frac{\mu_{\text{DR}}^2}{M_\phi^2} \int d^4x \sqrt{g} F_E^2. \quad (16)$$

Equation (14) agrees with the scalar logarithm in Eq. (2.12) and Appendix B of the full paper [2]. A direct flat-space check, extracting the factor e^2 from the polarization form factor, is

$$\Pi_{\log}(0) = \frac{1}{16\pi^2} \int_0^1 dx (1-2x)^2 \ln \frac{\Lambda^2}{M_\phi^2} = \frac{1}{48\pi^2} \ln \frac{\Lambda^2}{M_\phi^2}. \quad (17)$$

The determinant automatically combines the derivative-vertex bubble and the seagull. Their relative coefficient must be retained when reproducing (15) from propagators in Step 4.

The proper-time cutoff produces the additional constant $-\gamma_E$ in (10). The particular Pauli–Villars regulator used in the paper has a different finite convention. Agreement of the logarithmic slopes is exact; identifying the two regulated actions including their constants would require a finite conversion. In the paper’s Pauli–Villars normalization, a useful local reference prescription is

$$\delta_{3,\text{PV-!MS}}(\mu_R) = -\frac{e^2}{48\pi^2} \ln \frac{\Lambda^2}{\mu_R^2}. \quad (18)$$

This fixes a local subtraction scale. It does not impose a condition on the finite response to an external electric field.

5 The spacetime volume in the project calculation

In cosmic time write

$$ds^2 = -dt^2 + a^2(t)d\mathbf{x}^2, \quad a(t) = a_\star e^{H(t-t_\star)}. \quad (19)$$

For comoving spatial volume V_3 and a finite interval,

$$V_4[t_0, t_1] = V_3 \int_{t_0}^{t_1} dt a^3(t) = \frac{V_3}{3H} [a^3(t_1) - a^3(t_0)], \quad (20)$$

$$= \frac{V_3 a^3(t_0)}{3H} [e^{3H(t_1-t_0)} - 1]. \quad (21)$$

The expression in the outline omits the lower endpoint and implicitly fixes $a(t_0) = 1$. For a purely electric background of constant physical magnitude, $F_L^2 = -2E^2$. Continuing (8) back to Lorentzian signature gives

$$\Gamma_{L,F^2}^{(1)} = -\frac{e^2 I_\Lambda}{192\pi^2} \int d^4x \sqrt{-g} F_L^2 = \frac{e^2 E^2 I_\Lambda}{96\pi^2} V_4, \quad \Gamma_{L,\text{ct}} = \frac{\delta_3 E^2}{2} V_4. \quad (22)$$

The common factor $E^2 V_4$ cancels. The counterterm is a coefficient of a local operator, so its determination never depended on the size of the integration region. A spacetime-dependent coefficient would instead generate a term involving $(\nabla_\mu \delta_3) F^{\mu\nu}$ in the field equation. No such coefficient is produced here.

6 What the finite logarithm can tell us

Set $q = M_\phi^2/H^2$. The paper's flat-kinematics calculation with the curvature mass shift can be written

$$\delta_{3,\text{flat}+R}(2H^2) = \frac{e^2}{144\pi^2} \left[-3 \ln \frac{\Lambda^2}{H^2} + C_{\text{flat}+R}(q) \right], \quad (23)$$

where

$$C_{\text{flat}+R}(q) = 3 \ln q - 12q - 8 + 6(2q+1)^{3/2} \operatorname{arccoth} \sqrt{2q+1}. \quad (24)$$

The two logarithms in (23) reproduce (14). The remaining terms require a finite momentum-dependent form factor. For example,

$$\operatorname{arccoth} \sqrt{1+2q} = \frac{1}{2} \ln \frac{2}{q} + O(q), \quad 3 \ln q + 6(1+2q)^{3/2} \operatorname{arccoth} \sqrt{1+2q} = 3 \ln 2 + O(q \ln q). \quad (25)$$

This cancellation of the small-mass logarithm is invisible to the isolated b_2 term. It belongs to the finite flat-space form factor evaluated at a nonzero subtraction point; it does not establish the infrared behavior of the exact de Sitter response.

Higher heat-kernel coefficients give a local large-mass expansion in curvature and derivatives divided by M_ϕ^2 . On de Sitter, terms such as RF^2/M_ϕ^2 can look like further contributions to the coefficient of F^2 , but they are finite curvature terms, not a new four-dimensional UV divergence. When M_ϕ^2 is comparable to H^2 , retaining the full nonlocal kernel is essential.

7 Connection with the remaining calculation

The heat-kernel result resolves the volume question in the project and supplies the normalization of the local singularity in Step 4. For Hadamard states of the same scalar operator, the

ultraviolet counterterm coefficient is state independent. Finite response functions can still depend on the state and on its preparation.

There remains a finite freedom

$$\Gamma_{L,\text{ren}}[A] \longmapsto \Gamma_{L,\text{ren}}[A] - \frac{c}{4} \int d^4x \sqrt{-g} F^2, \quad (26)$$

which must be fixed by a renormalization condition [5]. In particular, exact de Sitter space does not diagonalize the two-point kernel by a global four-momentum. The notation $p^2 = 2H^2$ used in the paper’s approximation therefore needs a definition in the curved-space calculation. Step 4 compares a local reference subtraction with a finite condition on the retarded electric-field response. The latter is a possible prescription; the heat-kernel calculation does not select it.

References

- [1] *The photon two-point function and the scalar-loop counterterm in de Sitter space*, project notes, Secs. 1, 3–5.
- [2] M. Bastero-Gil, P. B. Ferraz, A. Torres Manso, L. Ubaldi and R. Vega-Morales, *Classical constant electric fields and the Schwinger effect in de Sitter*, arXiv:[2508.14973v2](#).
- [3] F. Bastianelli, J. M. Dávila and C. Schubert, *Gravitational corrections to the Euler–Heisenberg Lagrangian*, JHEP 03 (2009) 086, arXiv:[0812.4849](#).
- [4] D. V. Vassilevich, *Heat kernel expansion: user’s manual*, Phys. Rept. 388 (2003) 279–360, arXiv:[hep-th/0306138](#), Eq. (4.28).
- [5] S. Hollands and R. M. Wald, *On the Renormalization Group in Curved Spacetime*, Commun. Math. Phys. 237 (2003) 123–160, arXiv:[gr-qc/0209029](#).